Functional Annotation Report: MurA (Q88P88 / PP_0964) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

- **UniProt accession:** Q88P88
- **Gene:** *murA* (ordered locus **PP_0964**)
- **Organism:** *Pseudomonas putida* strain ATCC 47054 / DSM 6125 / KT2440 (PSEPK), a non-pathogenic soil/rhizosphere Gram-negative bacterium widely used as an industrial-biotechnology metabolic chassis (P 26913973).
- **Protein:** UDP-N-acetylglucosamine 1-carboxyvinyltransferase; EC **2.5.1.7**; synonyms **enoylpyruvate transferase / UDP-N-acetylglucosamine enolpyruvyl transferase (EPT)**.
- **Family / domains:** EPSP synthase family, **MurA subfamily** (HAMAP-Rule MF_00111); Pfam **PF00275 (EPSP_synthase)**; InterPro IPR005750 (UDP_GlcNAc_COvinyl_MurA), IPR001986/ IPR036968 (Enolpyruvate transferase domain/superfamily), IPR013792.

Verification outcome: The gene symbol *murA*, the EC 2.5.1.7 assignment, the EPSP-synthase/ MurA-subfamily classification, and the conserved enolpyruvyl-transferase domains are all mutually consistent and match a single, deeply conserved enzyme found across virtually all bacteria. This is **not** an ambiguous symbol: MurA is unambiguously the UDP-GlcNAc enolpyruvyl transferase of peptidoglycan biosynthesis. Because the enzyme is essential and universally conserved (HAMAP rule), the extensive mechanistic literature from *E. coli*, *Enterobacter cloacae*, and other bacteria applies directly to the *P. putida* KT2440 ortholog. Note: strain-specific biochemical papers on the KT2440 MurA are not available, so functional claims below are grounded in (i) the UniProt/HAMAP annotation of Q88P88 and (ii) primary mechanistic/structural studies of orthologous MurA enzymes.

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1. Summary (Answer to the Research Question)

MurA (Q88P88, PP_0964) is a soluble **cytoplasmic enzyme** that catalyzes the **first committed step of peptidoglycan (murein) biosynthesis**: the transfer of the intact **enolpyruvyl group from phosphoenolpyruvate (PEP) to the 3′-hydroxyl of UDP-N-acetylglucosamine (UDP-GlcNAc)**, yielding **enolpyruvyl-UDP-GlcNAc and inorganic phosphate** (EC 2.5.1.7) (P 40127436; 38532715). It is one of only two enzymes in nature (with EPSP synthase, AroA) known to perform enolpyruvyl transfer from PEP (P 40127436). The reaction proceeds by an **addition–elimination mechanism through a tetrahedral ketal intermediate**, mediated by an active-site cysteine and a large substrate-induced open→closed conformational change (P 9485407; 10694381; 9654090). MurA is **essential for bacterial viability**, is **feedback-inhibited** by the downstream product UDP-N-acetylmuramate, and is the molecular **target of the antibiotic fosfomycin** (P 12562791; 38532715; 40127436).

2. Primary Function: Reaction Catalyzed and Substrate Specificity

Reaction (EC 2.5.1.7):

> phosphoenolpyruvate + UDP-N-acetyl-α-D-glucosamine → phosphate + UDP-N-acetyl-3-O-(1-carboxyvinyl)-α-D-glucosamine (enolpyruvyl-UDP-GlcNAc)

MurA transfers the **entire enolpyruvyl moiety (C3 unit) of PEP** — not merely a phosphate — onto the **3′-OH of the GlcNAc sugar** of UDP-GlcNAc, releasing inorganic phosphate (P 38532715; 9485407). This is chemically unusual: PEP is normally used as a phosphoryl donor, but MurA (and EPSP synthase) instead cleave the C–O bond to transfer the enolpyruvyl group. "MurA and its related enolpyruvyl transferase, 5-enolpyruvyl-shikimate-3-phosphate (EPSP) synthase (AroA), are the only known enzymes to catalyze the unusual enolpyruvyl transfer from PEP" (P 40127436).

Substrate specificity: The two substrates are highly specific — **PEP** (the enolpyruvyl donor) and **UDP-GlcNAc** (the acceptor sugar nucleotide). The product enolpyruvyl-UDP-GlcNAc is the dedicated substrate of the next enzyme, MurB, which reduces it to UDP-N-acetylmuramate (UDP-MurNAc), the founding building block of the peptidoglycan stem peptide.

Catalytic mechanism (evidence from structure and stereochemistry): - The reaction is an **addition–elimination** proceeding through a **tetrahedral ketal intermediate**: the UDP-GlcNAc 3′-OH adds across the PEP double bond, then phosphate is eliminated (P 9485407).
- Crystallography of the *E. coli* **C115A mutant** trapped with a fluorinated tetrahedral-intermediate analogue defined the absolute configuration (2R) and showed addition occurs **to the 2-si face of PEP**, establishing the stereochemical course (P 9485407).
- An **active-site cysteine (Cys115** in *E. coli*/*Enterobacter* numbering) interacts directly with PEP and is central to catalysis and to protonation chemistry (P 10694381).

Induced-fit conformational cycle: - MurA is a two-domain protein that undergoes a large **open→closed transition upon substrate binding**. Small-angle X-ray scattering and fluorescence showed that "binding of UDP-GlcNAc to free enzyme results in substantial conformational changes, which can be interpreted as the transition from an open to a closed form" (P 9654090).
- A mobile **loop (residues 112–121) bearing the catalytic Cys** closes over the active site;

kinetic and X-ray studies of the Cys115Ser mutant established the role of this loop in the induced-fit mechanism (P 10694381).

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3. Subcellular Localization

MurA is a **soluble, cytoplasmic enzyme**. Peptidoglycan biosynthesis begins in the cytoplasm with a series of soluble Mur enzymes (MurA–MurF) that build the UDP-MurNAc-pentapeptide precursor before it is handed to membrane-associated steps (MraY, MurG) and finally polymerized in the periplasm. MurA specifically catalyzes the **cytoplasmic** enolpyruvyl-transfer step (P 38532715 describes it as "the peptidoglycan pathway cytoplasmic step"; P 35819545 calls MurA "an essential cytosolic enzyme in the biosynthesis of peptidoglycan"). Consistent with this, MurA carries no signal peptide or transmembrane segment and acts on soluble nucleotide-sugar substrates.

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4. Pathway Context and Biological Process

Pathway: UDP-N-acetylmuramate (peptidoglycan precursor) biosynthesis; more broadly, **bacterial cell-wall / murein biogenesis**.

- MurA performs **step 1** of the committed pathway, consuming **UDP-GlcNAc** (itself made from fructose-6-P via GlmS/GlmM/GlmU).
- Its product, **enolpyruvyl-UDP-GlcNAc**, is reduced by **MurB** (an FAD-dependent reductase) to **UDP-MurNAc**.
- MurC–MurF then sequentially add the pentapeptide (L-Ala, D-Glu, meso-DAP [in Gram-negatives such as *P. putida*], D-Ala-D-Ala) to form UDP-MurNAc-pentapeptide (Park's nucleotide).

- Downstream, MraY and MurG assemble lipid I/lipid II, which is flipped across the membrane and polymerized by penicillin-binding proteins into the peptidoglycan sacculus.

Thus MurA gates the flux of UDP-GlcNAc into cell-wall synthesis, sitting at the branch point between amino-sugar metabolism and peptidoglycan production. In *P. putida* KT2440, a Gram-negative bacterium with a thin peptidoglycan layer between inner and outer membranes, this pathway is required for cell-envelope integrity, cell division, and osmotic protection.

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5. Regulation

Feedback inhibition by the pathway product: UDP-N-acetylmuramate (UDP-MurNAc / UNAM) — produced two steps downstream by MurB — "binds tightly to MurA forming a dormant UNAM-PEP-MurA complex and acting as a MurA feedback inhibitor" (P 38532715). Molecular-dynamics analysis indicates UNAM stabilizes the **closed conformation** via hydrogen bonds to conserved arginines (Arg120, Arg91) and has a longer residence time than the forward substrate UDP-GlcNAc, providing product-level control that prevents over-commitment of UDP-GlcNAc to peptidoglycan (P 38532715). This couples the flux through MurA to the demand for cell-wall precursors.

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6. Essentiality, Drug-Target Status, and Fosfomycin

- **Essentiality:** A lethal ΔmurA deletion in *E. coli* is viable only when complemented by a functional *murA* gene, demonstrating that MurA is essential for bacterial growth (P 12562791).
- **Fosfomycin target:** Fosfomycin (a PEP-mimetic epoxide phosphonate) is "an irreversible covalent inhibitor of UDP-GlcNAc enolpyruvyl transferase (MurA)" that "binds to the active site of MurA in competition with substrate phosphoenolpyruvate (PEP) and undergoes the ring-opening nucleophilic attack of an active-site cysteine"

(P 40127436).
This covalent adduct on the catalytic Cys inactivates the enzyme, blocking cell-wall synthesis.

- **Resistance determinant:** Mutation of the active-site cysteine (e.g., **Cys→Asp**) confers intrinsic fosfomycin resistance; *Chlamydia* MurA naturally carries this substitution and is fosfomycin-resistant, imparting resistance when expressed in *E. coli* (P 12562791).
- **Drug-discovery relevance:** MurA has no mammalian homolog and is a "validated target enzyme for antibacterial drug discovery," motivating ongoing development of covalent (e.g., chloroacetamide-warhead) and natural-product inhibitors (P 36126388; 35819545). This is contextual to the enzyme family and not specific to *P. putida* susceptibility.

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7. Evidence Basis and Confidence

Claim	Type of evidence	Strength
Reaction/substrates/products (EC 2.5.1.7)	Biochemical + structural (orthologs) + UniProt/HAMAP annotation of Q88P88	High
Cytoplasmic localization	Pathway biochemistry; absence of signal/TM segments; family annotation	High
Addition–elimination via tetrahedral ketal; stereochemistry	X-ray of trapped intermediate analogue (E. coli C115A) (P 9485407)	High (ortholog)
Active-site Cys, induced-fit loop closure	Kinetics, X-ray, SAXS (Enterobacter/E. coli) (P 10694381 P 9654090)	High (ortholog)
Product feedback inhibition (UDP-MurNAc)	MD + prior biochemistry (P 38532715)	Moderate–High
Essentiality	Genetic complementation (P 12562791)	High (ortholog); MurA universally essential
Fosfomycin inhibition/ resistance	Structural/pharmacological (P 40127436 P 12562791)	High (family-level)

Inference for Q88P88 specifically: Because MurA is governed by HAMAP rule MF_00111 and the *P. putida* KT2440 sequence is a bona fide MurA-subfamily member with the conserved catalytic Cys and Arg residues, the mechanistic, localization, and regulatory conclusions transfer with high confidence to PP_0964. Direct enzymology on the KT2440 protein itself has not been reported.

8. Supported and Refuted Hypotheses

Supported: - H1: Q88P88 is a UDP-GlcNAc enolpyruvyl transferase catalyzing PEP + UDP-GlcNAc → enolpyruvyl-UDP-GlcNAc + Pi. ✔ - H2: The enzyme is cytoplasmic and functions in the first committed step of peptidoglycan biosynthesis. ✔ - H3: Catalysis uses an active-site cysteine and an induced-fit tetrahedral-intermediate mechanism. ✔ - H4: MurA is essential and is the fosfomycin target; the catalytic Cys governs drug sensitivity. ✔ - H5: MurA is feedback-inhibited by the downstream product UDP-MurNAc. ✔ (computational/biochemical)

Refuted / excluded: - The gene symbol is **not** ambiguous; there is no competing well-supported alternative function for this locus. MurA is not a transporter, structural protein, or signaling molecule — it is a biosynthetic transferase.

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9. Limitations and Future Directions

- No *P. putida* KT2440-specific enzymological, structural, or knockout study of MurA was found; conclusions rest on orthologs and family-level annotation.
- The precise residue numbering (catalytic Cys, regulatory Arg) in Q88P88 should be confirmed by sequence alignment to the *E. coli*/*Enterobacter* references before residue-level claims are made for KT2440.
- Whether *P. putida* KT2440 has a second MurA paralog or additional envelope-stress regulation of *murA* expression is not established here.
- Future work: recombinant KT2440 MurA kinetics with PEP/UDP-GlcNAc; fosfomycin MIC and Cys-residue identity; structural confirmation of the closed complex.

Key References

- P 40127436
  — Kim & Lees (2025), molecular pharmacology of fosfomycin/MurA.
- P 9485407
  — Skarzynski et al. (1998), tetrahedral-intermediate structure & stereochemistry.

- P 10694381
  — Schönbrunn et al. (2000), Cys115 loop / induced fit.
- P 9654090
  — Schönbrunn et al. (1998), open→closed conformational change (SAXS).
- P 38532715
  — de Oliveira et al. (2024), UDP-MurNAc feedback inhibition (MD).
- P 12562791
  — McCoy et al. (2003), essentiality & fosfomycin resistance determinant.
- P 35819545 /
  36126388 — MurA as an essential cytosolic drug target.
- P 26913973
  — Belda et al. (2016), *P. putida* KT2440 genome/chassis.